ASSIGNMENT 2

Applied Concepts and Objectives of Cryptography

Questions with Answers | Short answers: 2–3 sentences | Long answers: detailed explanations

Q1. (Short) List the four fundamental cryptographic goals/objectives.

Answer: The four fundamental objectives are **Confidentiality, Integrity, Authentication,** and **Non-repudiation**. Together, they help keep information secret from unauthorized people, protect it from changes, verify who is communicating, and provide evidence that a person performed an action or sent a message.

Q2. (Short) Define “Confidentiality” as a cryptographic goal with one example.

Answer: Confidentiality means that information can be read only by authorized people. For example, when a banking app encrypts account details while they travel across the internet, an outsider who intercepts the data should not be able to understand it without the proper key.

Q3. (Long) Explain each cryptographic objective in detail with real-world scenarios: Confidentiality, Integrity, Authentication, and Non-repudiation. Why is it important to achieve all four together in a secure communication system?

Answer: 1. Confidentiality: Confidentiality protects information from being seen by unauthorized people. Encryption changes readable data (plaintext) into unreadable data (ciphertext), which can be restored only by someone with the correct decryption key. *Scenario:* HTTPS encrypts information sent between a customer’s browser and an online shopping website, helping protect passwords and payment details from eavesdroppers.

2. Integrity: Integrity ensures that data has not been changed, deleted, or damaged without authorization. Cryptographic hash functions and message authentication codes (MACs) can help a receiver detect changes. *Scenario:* A software provider publishes a file hash so a user can check that the downloaded installer matches the original. A plain hash alone does not prove who published the file unless it is obtained through a trusted channel or digitally signed.

3. Authentication: Authentication verifies the identity of a user, device, or server, or confirms the origin of a message. Digital certificates, digital signatures, passwords combined with additional checks, and challenge-response protocols can support authentication. *Scenario:* During an HTTPS connection, a browser checks the website’s digital certificate to help confirm that it is communicating with the intended website.

4. Non-repudiation: Non-repudiation provides evidence that can help prevent a sender from credibly denying an action, such as signing a particular digital document. A digital signature created with a person’s private key can be verified using the corresponding public key. Its legal or practical strength also depends on protecting the private key, reliable identity checks, system records, and applicable rules. *Scenario:* An organization digitally signs a contract so the recipient can verify that the document has not changed and that the signature corresponds to the signer’s public key.

Why all four matter: Encryption alone may hide a message but does not necessarily prove who sent it or whether it was changed. Integrity without authentication may leave uncertainty about who created the data, while authentication without confidentiality may expose private information. Combining the four goals provides stronger protection for privacy, accuracy, identity, and accountability in systems such as online banking, e-commerce, and secure business communication.

Q4. (Short) What is the difference between a substitution cipher and a transposition cipher? Give one example of each.

Answer: A **substitution cipher** replaces each letter or symbol with another symbol, while a **transposition cipher** rearranges the positions of the original letters without changing the letters themselves. For example, a Caesar cipher is a substitution cipher (HELLO becomes KHOOR with a shift of 3); a rail-fence cipher is a transposition cipher.

Q5. (Long) Discuss the evolution from classical ciphers to modern cryptographic standards (DES → AES, RSA, SHA). Highlight key milestones and motivations for change.

Answer: Classical ciphers: Early systems used manual methods such as Caesar substitution and transposition. Later, polyalphabetic methods such as the Vigenère cipher made simple letter-frequency analysis more difficult. These systems were useful historically but became vulnerable to improved analysis and computing.

Mechanical and electromechanical era: Devices such as rotor machines, including the Enigma machine, automated more complex transformations. Their security depended on operating procedures and key settings, and weaknesses in implementation or key handling could be exploited.

Computer-era symmetric encryption and DES: With digital computers, cryptography moved toward precise algorithms that process bits. The U.S. Data Encryption Standard (DES) was adopted in 1977. DES uses a 56-bit effective key; as computing power increased, exhaustive key-search attacks became increasingly practical. Triple DES (3DES) extended DES by applying it multiple times, but it was slower and is now being phased out in many uses.

Public-key cryptography and RSA: In 1976, Whitfield Diffie and Martin Hellman published the idea of public-key cryptography and key exchange. RSA, introduced in 1977, became a major public-key algorithm. Unlike symmetric encryption, public-key systems use related public and private keys, helping solve the problem of securely sharing a secret key in advance. RSA is often used for signatures or key establishment rather than encrypting large amounts of data directly.

SHA hash standards: Secure Hash Algorithms (SHA) produce fixed-length message digests used to help detect changes and support digital signatures. SHA-1 was widely used but is no longer considered secure against collision attacks. SHA-2 and SHA-3 provide stronger modern options for many applications. A hash is not encryption: it does not use a decryption key to recover the original message.

AES replaces DES: The Advanced Encryption Standard (AES) was selected by NIST in 2001 to replace DES. AES supports 128-, 192-, and 256-bit keys and is widely used for fast symmetric encryption. Modern systems often combine algorithms: public-key cryptography or key exchange helps establish keys, AES encrypts the actual data, and authenticated-encryption modes such as AES-GCM provide confidentiality and integrity together.

Main reasons for change: The main motivations were growing computing power, cryptanalytic discoveries, the need for stronger keys, efficient digital communication, secure key exchange, and the need to protect data and verify its integrity in networked systems.

Q6. (Short) What is a “key” in cryptography, and why is key management important?

Answer: A **cryptographic key** is a value used by an algorithm to encrypt, decrypt, sign, or verify information. **Key management** is important because keys must be generated securely, stored safely, shared only with authorized parties, rotated when appropriate, and revoked or replaced if compromised; a strong algorithm cannot protect data if its key is exposed.

Q7. (Long) Cryptography plays a vital role in securing computer networks. Explain, with suitable examples, how cryptographic techniques are used to protect data during transmission, prevent unauthorized access, and detect tampering in a network.

Answer: 1. Protecting data during transmission: Encryption converts readable information into ciphertext. In HTTPS/TLS, a client and server use a secure handshake to authenticate the server and establish session keys; symmetric encryption then protects the data sent during the connection. *Example:* When a student logs in to a school portal, TLS helps stop people on the same public Wi-Fi network from reading the login details in transit.

2. Preventing unauthorized access: Cryptography supports identity checks and access control. Passwords should be stored as salted password hashes using a suitable password-hashing function, rather than as readable text. Multi-factor authentication and cryptographic challenge-response methods can make account access harder for an attacker. Digital certificates can also help devices verify servers or one another.

3. Detecting tampering: Cryptographic hashes produce a digest for data, allowing a receiver to compare the received content with a trusted digest. A MAC or authenticated-encryption scheme can detect unauthorized changes and, when properly configured, authenticate the data. Digital signatures can provide evidence of origin and integrity. *Example:* A network service can use a message authentication code to reject a message whose contents were altered in transit.

Simple secure-communication flow:

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Sender → Encrypt / authenticate → Network → Verify / decrypt → Receiver
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In practice, secure networks combine these methods rather than relying on a single technique. Encryption protects secrecy, authentication checks identities or message origins, and integrity checks help reveal unauthorized changes. Secure key storage, certificate validation, software updates, and correct configuration are also essential.

Q8. (Short) Differentiate between a passive attack and an active attack on a cryptographic system, with one example of each.

Answer: A **passive attack** observes or monitors communication without changing the data; an example is eavesdropping on unencrypted network traffic. An **active attack** changes, inserts, deletes, or disrupts data; an example is a man-in-the-middle attacker altering a message while it travels between two parties.

Q9. (Long) Describe how cryptographic techniques evolved historically to counter emerging security threats. Include a discussion of the shift from symmetric-only systems to the introduction of asymmetric (public-key) cryptography, and how this addressed the key-distribution problem.

Answer: Early cryptography mainly used substitution and transposition methods. Their security relied on keeping the method or secret arrangement hidden, but language patterns and later statistical analysis could reveal messages. Mechanical rotor machines increased complexity, yet their security could still be undermined by weaknesses in the machine, operating procedures, or key settings.

With electronic computers, algorithms became more systematic and capable of processing large amounts of digital data. Symmetric systems such as DES offered efficient encryption, but the sender and receiver needed to share the same secret key securely. If the key was sent over an insecure channel or exposed, an attacker could read messages protected by it. Distributing separate secret keys to many users also became difficult as networks grew.

In 1976, Diffie and Hellman introduced a landmark public-key approach, and RSA followed in 1977. In asymmetric cryptography, each user has a public key that can be shared and a private key that must be kept secret. Public-key encryption can allow a sender to protect information for the intended recipient, while digital signatures can be created with a private key and verified with a public key. Key-agreement protocols, such as Diffie–Hellman, allow parties to establish a shared secret over a public channel, although authentication is needed to prevent impersonation or a man-in-the-middle attack.

Modern systems commonly use a **hybrid approach**: public-key cryptography or authenticated key exchange establishes a temporary symmetric session key, and a fast symmetric algorithm such as AES protects the bulk data. Hash functions such as SHA-2 or SHA-3 help with integrity-related tasks and digital-signature schemes. Newer protocols, stronger key sizes, authenticated encryption, certificate systems, and better key-management practices respond to increasing computing power and new attack techniques. Thus, cryptography evolved from simple secret codes into layered systems designed for large-scale, authenticated digital communication.

Q10. (Short) Give one example each of a modern symmetric algorithm and a modern asymmetric algorithm used today.

Answer: A modern **symmetric algorithm** is **AES (Advanced Encryption Standard)**, which uses the same secret key for encryption and decryption. A modern **asymmetric algorithm** is **RSA**, which uses a public/private key pair; elliptic-curve algorithms are also widely used today.

Q11. (Long) Discuss the classification of cryptography into Classical Cryptography and Modern Cryptography. Explain the transition from manual/mechanical methods to computer-based algorithms, mentioning key milestones and technologies involved.

Answer: 1. Classical cryptography: Classical cryptography includes early methods that usually worked on letters or symbols. *Substitution ciphers* replace symbols (for example, the Caesar cipher), while *transposition ciphers* rearrange their order (for example, the rail-fence cipher). The Vigenère cipher used changing substitution patterns. These methods could be performed by hand and were important before widespread computing, but many are vulnerable to frequency analysis and other forms of cryptanalysis.

2. Mechanical and electromechanical methods: As communication became more complex, machines such as rotor ciphers automated substitutions and produced many possible transformations. The Enigma machine is a well-known historical example. This period showed that machine complexity alone does not guarantee security; key settings, operator practices, and weaknesses in the design also matter.

3. Modern cryptography: Modern cryptography is based on mathematical algorithms designed for digital data and evaluated against known attack models. It includes symmetric encryption (DES historically and AES today), asymmetric algorithms and key exchange (such as RSA and Diffie–Hellman), cryptographic hash functions (such as SHA-2 and SHA-3), message authentication codes, and digital signatures. These techniques are implemented in software and hardware and are used in protocols such as TLS, secure messaging, VPNs, and digital payment systems.

Important milestones:
• **1940s:** Development of information theory and major advances in computer-based cryptanalysis.
• **1970s:** DES standardized in 1977; Diffie–Hellman public-key ideas published in 1976; RSA introduced in 1977.
• **1990s–2000s:** Internet and e-commerce growth increased the need for secure web connections; SHA hash standards and digital certificates became important parts of security systems.
• **2001:** AES was selected as the successor to DES for modern symmetric encryption.
• **2010s onward:** SHA-3 was standardized, authenticated-encryption methods became common, and cryptographic systems continued to adapt to new attacks and computing capabilities.

Key difference: Classical methods were often manual or mechanical and focused on transforming letters. Modern cryptography uses formal mathematical algorithms, binary data, computational security, large keys, and carefully managed keys. Modern security also depends on correct implementation, secure protocols, authentication, and regular updates—not just on choosing a strong algorithm.